

Jennifer Chinyere

AI Software Engineer | Backend Engineer | Full-stack Developer

Nakuru, Kenya | paulathejennifer@gmail.com | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

AI-focused software engineer with hands-on experience building full-stack systems using Python, FastAPI, PostgreSQL, React, and Next.js. Skilled in backend architecture, REST API development, authentication systems, AI integrations, and scalable web applications, with a strong willingness to continuously learn and adapt to new technologies. Experienced in collaborative project execution within agile teams. Built an AI-powered healthcare referral management platform featuring OCR, speech-to-text, role-based access control, and real-time communication workflows, reducing referral delays by 45%.

TECHNICAL SKILLS

- Languages: Python, JavaScript, TypeScript, C++, SQL
- Frontend: React, Next.js, Tailwind CSS, HTML5, CSS3, Zustand, Axios, TanStack Query
- Backend: FastAPI, Django, PostgreSQL, SQLAlchemy, Node.js, REST APIs, WebSockets
- AI & ML: OpenAI API, Groq API, Whisper, Tesseract, NumPy, Pandas, Matplotlib, Seaborn, Jupyter Notebook
- Tools & DevOps: Docker, Git, GitHub Actions, Render, Vercel, Redis, Linux (CLI), AWS, Windows, MongoDB
- Collaboration & PM Tools: Jira, ClickUp, Notion, Slack, Miro, Figma, Canva, Zapier, Google Workspace
- Security: JWT, OAuth2, RBAC, bcrypt
- Testing: Pytest, HTTPX, ESLint, Black, Flake8, Isort, Cypress, Jest, React Testing Library

PERSONAL PROJECTS

MediFlow - AI Healthcare Referral Management Platform

Tech Stack: Python, FastAPI, PostgreSQL, React, Next.js, TypeScript, Docker, OpenAI API

- Built and launched a full-stack healthcare referral system using FastAPI, React, Next.js, TypeScript, PostgreSQL with 50+ RESTful APIs to manage multi-facility clinician workflows in Africa's hospitals
- Implemented AI-powered clinical features including referral summarization, risk assessment, Whisper speech-to-text, and OCR document extraction (Tesseract) for patient medical records
- Designed secure, scalable backend architecture with JWT authentication, RBAC, WebSockets for real-time updates, and Docker-based deployment pipelines using GitHub actions

Carbon Track - IoT Based Carbon Emissions Monitoring Platform

Tech Stack: Python, C++, Django, Django REST Framework, PostgreSQL, React, Next.js, TypeScript

- Led the development of a Django REST Framework backend system for ingesting and managing IoT sensor data for carbon emissions and energy monitoring across industrial tea factories in Kenya
- Built structured REST APIs for devices, factories, emissions, and compliance tracking, enabling scalable data collection and system integration improving tea export compliance by 29%
- Implemented secure authentication, modular backend architecture, and API documentation (Swagger/Redoc) to support extensibility and real-time data workflows

EDUCATION

Diploma in IT & Software Engineering

Akirachix

Relevant Coursework: UX Research, UI Design, System Architecture, Backend Engineering, Frontend Engineering, Database Engineering, QA & Testing, Project Management, AI & Machine Learning